

Inception: Efficiently Computable Misinformation Attacks on Markov Games

Motivation

How can misinformation effect games?

Original			Influenced		
	L	R		L	R
U	0, 5	1, 0	U	0, 5	1, 6
D	1, ϵ	0, 0	D	1, ϵ	0, 1
S	1, 0	0, ϵ	S	1, 0	0, ϵ

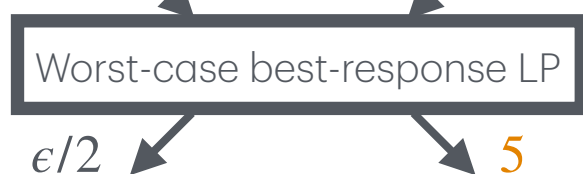
Original: P1 plays D, so P2 gets 0

Influenced: P1 plays U, so P2 can get 5

Attack Algorithms

Idea: make P1 think P2 plays j

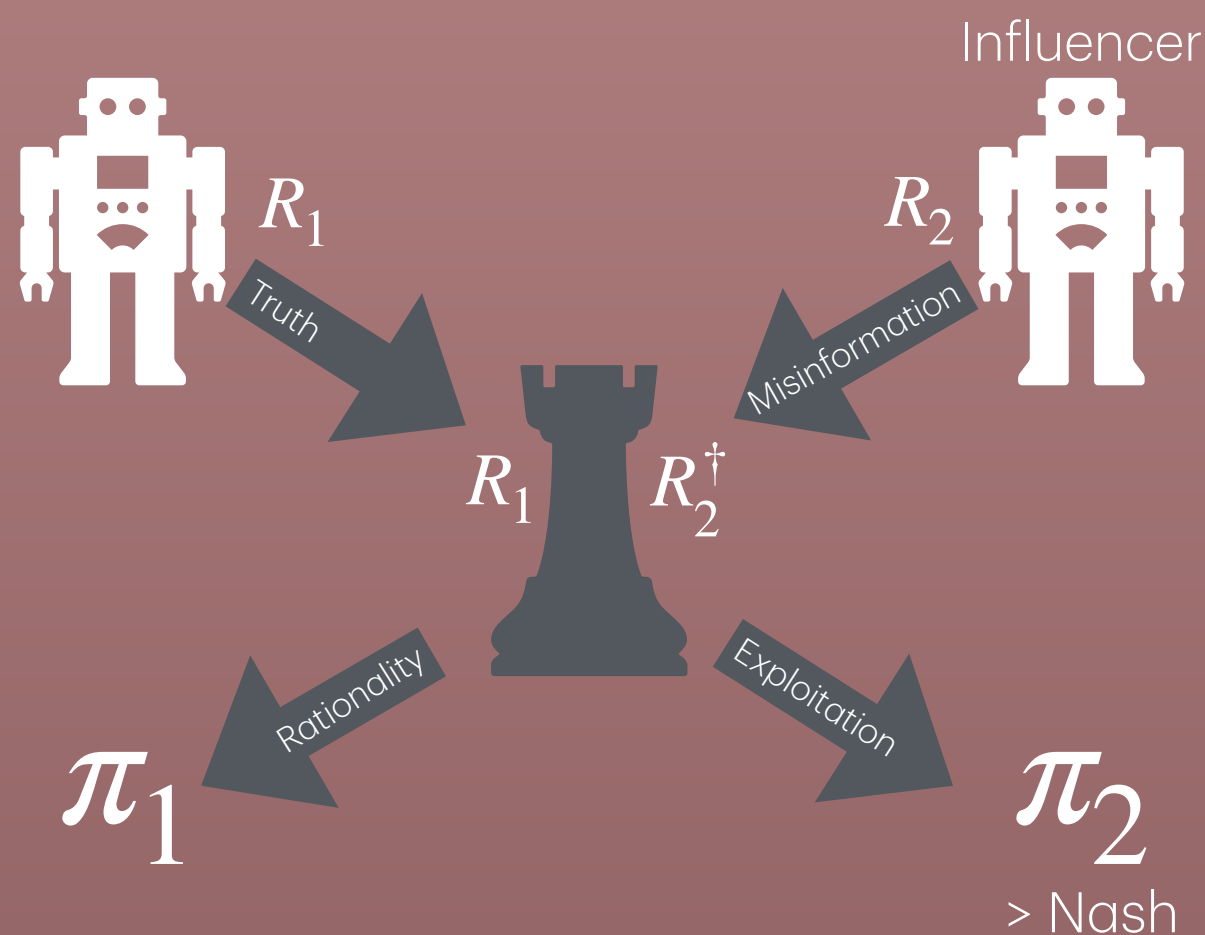
$j = L$			$j = R$		
	L	R		L	R
U	0, 5	1, 0	U	0, 5	1, 6
D	1, ϵ	0, 0	D	1, ϵ	0, 1
S	1, 1	0, ϵ	S	1, 0	0, ϵ



Results

- **Worst-case best-responses** are poly-time computable for any finitely generated belief set.
- Optimal pure policy-dominant **misinformation attacks** are poly-time computable.

A player can spread reward **misinformation** to exploit **rational** opponents.



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